BEFORE THE PUBLIC UTILITIES COMMISSION OF THE STATE OF CALIFORNIA

Application of Pacific Gas and Electric Company (U39M) for Approval of its Proposal for a Day-Ahead Real Time Rate and Pilot to Evaluate Customer Understanding and Supporting Technology.	Application 20-10-011 (Filed October 23, 2020)

RESPONSE OF THE VEHICLE-GRID INTEGRATION COUNCIL TO PACIFIC GAS AND ELECTRIC'S PETITION FOR MODIFICATION OF D. 21-11-017

Zach Woogen
Executive Director
Vehicle-Grid Integration Council
1401 21st St., Suite 5409
Sacramento, California 95811
(279) 895-7855
vgicregulatory@vgicouncil.org

Grace Pratt
Caliber Strategies
801 K Street, Suite 2800
Sacramento, CA 95814
(818) 264-6437
grace@caliberstrat.com

March 7, 2025

BEFORE THE PUBLIC UTILITIES COMMISSION OF THE STATE OF CALIFORNIA

Application of Pacific Gas and Electric Company (U39M) for Approval of its Proposal for a Day-Ahead Real Time Rate and Pilot to Evaluate Customer Understanding and Supporting Technology.	Application 20-10-011 (Filed October 23, 2020)

RESPONSE OF THE VEHICLE GRID INTEGRATION COUNCIL TO PACIFIC GAS AND ELECTRIC'S PETITION FOR MODIFICATION OF D. 21-11-017

Pursuant to Rule 16.4(f) of the Rules of Practice and Procedure of the California Public Utilities Commission ("Commission"), the Vehicle-Grid Integration Council ("VGIC") responds to Pacific Gas and Electric's ("PG&E") *Petition for Modification of D.21-11-017*, filed on February 5, 2025.

I. INTRODUCTION.

VGIC is disappointed by PG&E's Petition for Modification ("PFM") to terminate implementation efforts of its Day-Ahead Hourly Real-Time Pricing Business Electric Vehicle ("DAHRTP-BEV") rate and associated export compensation pilot for non-Net Energy Metering ("NEM") customers. In its below response, VGIC details an alternative path forward in recognition of the concerns raised by PG&E and the present landscape of dynamic and export rate opportunities available to commercial electric vehicle ("EV") charging customers that aim to support the grid and lower their total cost of ownership.

The final DAHRTP-BEV rate represents groundbreaking progress in the landscape of real-time pricing ("RTP") rates and commercial electric vehicle ("EV") managed charging. DAHRTP-BEV was approved not as merely another pilot, as initially proposed by PG&E. Instead, the Commission unanimously and decisively directed PG&E to implement DAHRTP-BEV as a full-scale optional real-time pricing ("RTP") rate, the first of its kind in the nation. When approved, D.21-11-017 and the associated D.22-10-024 sent a powerful, unwavering signal to customers and solution providers seeking opportunities to manage commercial EV charging load and implement vehicle-to-grid ("V2G") export: California is committed to unlocking the benefits of this technology. Acting on this regulatory certainty and finality, commercial EV charging site hosts, EV fleets, project developers and financing entities, and technology providers mobilized in preparation for the anticipated October 2023 launch of DAHRTP-BEV and the associated non-NEM export pilot. In the months and years that followed, PG&E unexpectedly delayed implementation first by six months, then by a year, and then by two years, and has now filed the PFM to request approval to terminate implementation efforts entirely. Meanwhile, California ratepayers are in dire need of cost relief and the grid demands new sources of low-cost load flexibility to support reliability during emergencies. With a majority of EV charging use cases characterized by long dwell times that bring inherent load flexibility and the potential for power export, VGIC views ending DAHRTP-BEV and the associated export pilot before it begins as a significant misalignment between the needs of California ratepayers and the opportunity presented by the fast growing network of connected, flexible commercial EV charging sites. DAHRTP-BEV exists within an increasingly small set of strategies that customers and EV service providers ("EVSP") can use in California to unlock net peak load reductions and alleviate distribution system constraints, and, in turn, mitigate further upward pressure on rates.

Approving the PFM as filed would eliminate the benefits of DAHRTP-BEV and the associated export pilot without parties having had the chance to collect evidence to understand the scale of these benefits. Furthermore, approving the PFM would also severely undermine the regulatory certainty typically broadcast from Commission decisions and inhibit market development for emerging solutions that can yield substantial net benefit for California ratepayers.

With this in mind, VGIC acknowledges that several of the implementation challenges PG&E cites in its PFM do present barriers to launching DAHRTP-BEV and the associated export pilot as envisioned. In its below response, VGIC offers a reasonable, no-regrets path forward that we believe achieves a compromise between PG&E's cited concerns over implementing DAHRTP-BEV as directed, the existing ratepayer cost pressures top of mind for customers, the Public Advocate's Office, and the Commission, and the imperative to support the growth of a robust, self-sustaining managed and integrated EV charging market that yields net benefits for all parties involved. Specifically, VGIC offers the following comments in response to PG&E's PFM:

- The Commission should deny the PFM as filed because the requested relief is insufficiently supported by the record in the proceeding or other matters officially noticed.
    - The PFM erroneously asserts Commission direction on RTP rates.
    - The PFM would result in discriminatory treatment toward certain customers.
    - The PFM contains unsupported claims regarding the prioritization of billing system updates.
    - The PFM violates Commission Rules by seeking to modify two Commission decisions through a single PFM.
    - The PFM misrepresents the state of PG&E's Vehicle-Grid Integration ("VGI") Pilots.
    - The PFM misrepresents BEV Parties' positions.
- The Commission should deny the PFM as filed because the requested relief undermines the Commission's objectives and principles and would materially harm PG&E ratepayers.
    - The PFM would materially harm customers and solution providers working to unlock load flexibility and exports to support California's grid.
    - The Commission has staked the future of load flexibility and distributed exports on a single strategy, CalFUSE, for which available data suggests there is little customer and vendor interest and limited load shift results.

- Over-relying on a single approach is at odds with established energy planning principles of resource diversity and supplier diversity, strategies intended to mitigate risk.
- Approving the PFM would send a strong signal that regulated utilities can successfully delay and then terminate implementation efforts for Commission orders they do not wish to comply with.

- VGIC proposes Alternative Proposed Modifications for the Commission's consideration.
  - The Commission should direct PG&E to leverage its **already completed work** on dynamic distribution components to promote compliance with the California Energy Commission's Load Management Standard.
  - The Commission should direct PG&E to **enable shadow billing for initial DAHRTP-BEV implementation.**
  - The Commission should authorize budget transfer within the **already approved DAHRTP-BEV funds** to support the Alternative Proposed Modifications.

II. THE COMMISSION SHOULD DENY THE PFM AS FILED BECAUSE THE REQUESTED RELIEF IS INSUFFICIENTLY SUPPORTED BY THE RECORD IN THE PROCEEDING OR OTHER MATTERS OFFICIALLY NOTICED.

a. The PFM erroneously asserts Commission direction on RTP rates.

In the PFM, PG&E asserts that "the rate structure authorized in D.21-11-017 has become outdated in that it does not meet the current guidance in the Energy Division CalFUSE guidance framework (CalFUSE or CalFUSE framework)…"[1] VGIC disagrees that this offers support for the requested relief as the Energy Division's CalFUSE framework is not equivalent to Commission direction. Notably, the Commission has not directed PG&E to adopt the CalFUSE framework for business EV ("BEV") customers. In fact, the only Commission guidance on this matter is found in D.24-01-032, in which the Commission denies PG&E's request to implement the CalFUSE framework for BEV customers, an occurrence detailed by PG&E in the

[1] PFM at 2.

PFM.[2] VGIC believes the PFM conflates Energy Division guidance with Commission direction and, as a result, the Commission should disregard the use of the CalFUSE framework's existence as evidence to support the PFM's request.

b. The PFM would result in discriminatory treatment toward certain customers.

The PFM critically fails to consider the impacts that terminating DAHRTP-BEV implementation and the associated non-NEM export pilot would have on certain BEV customers. While PG&E notes that both bidirectional and unidirectional charging BEV customers are currently eligible to participate in PG&E's VGI RTP rate implemented through its vehicle-to-everything ("V2X") Pilots, these pilots have several restrictions that differentiate the opportunity from DAHRTP-BEV and the associated non-NEM export pilot. First, the V2X Pilots are operating under budget limitations, including for program administration and equipment rebates, which will likely mean the V2X Pilot will, at some point, come to an end. Specifically, PG&E's website notes there are only incentives remaining for 126 chargers.[3] Second, PG&E's V2X Pilots restricts the customer pool by imposing an additional requirement for customers: they must participate through a PG&E-approved Automated Service Provider ("ASP"). In contrast, DAHRTP-BEV does not require enrollment through a PG&E-approved ASP. Third, for bidirectional charging customers, only customers with equipment listed on the "Eligible Product List for the V2X Commercial Pilot" may participate in the V2X Pilot. PG&E's website notes there are only three eligible vehicles and four eligible chargers for the V2X Commercial Pilot.[4] Meanwhile, VGIC

[2] PFM at 13-14.
[3] PG&E. *Vehicle-to-Everything (V2X) pilot program.* https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html. (Accessed March 7, 2025).
[4] *Ibid.*

notes that all school buses sold in California under the California Air Resources Board's ("CARB") Hybrid and Zero-Emission Truck and Bus Voucher Incentive Project ("HVIP") or Zero-Emission School Bus and Infrastructure ("ZESBI") programs must be capable of bidirectional charging.[5] As for available chargers, VGIC notes that other manufacturers, including Fermata Energy, InCharge, Heliox, Ford Motor Company, GM Energy, Tesla, dcbel, and Wallbox offer bidirectional capable charging equipment that can interconnect under Rule 21 or via non-Rule 21 temporary interconnection pilots.[6] While some of these chargers are currently marketed toward residential customers, VGIC anticipates a share of residential chargers may soon be used to support light-duty fleet V2G.

These three restrictions results in discriminatory treatment toward BEV customers that do not wish to participate in the VGI RTP over concerns that the pilot may soon exhaust authorized funds or the inability to procure equipment eligible for the V2X Pilots. VGIC recommends the Commission consider the impacts the PFM would have on these customers.

c. **<u>The PFM contains unsupported claims regarding the prioritization of billing system updates.</u>**

The PFM details that one of the three developments that have taken place since the Commission approved the DAHRTP-BEV rate as follows:

> "delays in PG&E's billing system upgrade are preventing PG&E from implementing DAHRTP BEV in the new Billing Cloud Service (BCS) system when it launches, without causing further delays in the

[5] *See* CARB. *Implementation Manual for the Hybrid and Zero-Emission Truck and Bus Voucher Incentive Project.* https://californiahvip.org/wp-content/uploads/2024/10/FY23-24-HVIP-Implementation-Manual-103124.pdf. October 31, 2024; CARB *Implementation Manual Appendix G: Zero Emission School Bus & Infrastructure Incentives*. https://californiahvip.org/wp-content/uploads/2024/04/Final-FY23-24-ZESBI-Implementation-Manual.pdf. April 30, 2024.

[6] A.24-03-019. *Direct Testimony of Zach Woogen on Behalf of the Vehicle-Grid Integration Council*. January 8. 2025. Pages 6-10. https://docs.cpuc.ca.gov/PublishedDocs/SupDoc/A2403019/7930/553184229.pdf

> implementation of other rates that the Commission has approved that PG&E believes are of higher priority than DAHRTP BEV."[7]

Furthermore, the Deceleration of Emily Bartman notes that the Net Billing Tariff (NBT) and the Residential Income Graduated Fixed Charge "have a much greater potential to impact affordability, as NBT will reduce cost-shift and the residential fixed charge will reduce bills for low income customers."[8] The Declaration

includes a referencing detailing that NBT implementation should occur by June 2024 and Residential Fixed Charges should be implemented by March 31, 2026. These dates were determined by Decisions authorized in December 2022 and May 2024, respectively. In contrast, DAHRTP-BEV implementation was authorized in November 2021 and slated for October 2023 launch. This timeline predates both NBT and Residential Fixed Charges decisions and target implementation dates.

The existing timelines suggest a "first come, first serve" approach that would position DAHRTP-BEV for timelier implementation. Given the Commission has not issued direction detailing the relative prioritization of these initiatives, PG&E's argument based on its internal assessment on the relative importance of these initiatives should be disregarded as support for the PFM.

With that in mind, VGIC strongly urges the Commission to adopt a prioritization to guide PG&E's initiatives, which may be a "first come, first serve" approach, or may be based on analysis of potential ratepayer benefit, ability to achieve state goals, or other Commission objectives.

---

[7] PFM at 1.

[8] Declaration of Emily Bartman at 3.

d. The PFM violates Commission Rules by seeking to modify two Commission decisions through a single PFM.

Through the PFM, PG&E petitions to modify not only D.21-11-017, but also notes the following:

> "Because the non-NEM export compensation pilot authorized by D.22-10-024 is dependent on the underlying DAHRTP BEV rate authorized by D.21-11-017, eliminating the DAHRTP BEV rate would also moot the non-NEM export pilot."[9]

VGIC appreciates PG&E's acknowledgement that modification of D.21-11-017 would materially impact the Commission direction adopted in D.22-10-024, making the latter "moot." Section VIII of the PFM reiterates that "With the elimination of DAHRTP BEV as an optional rate, the export compensation pilot authorized by D.22-10-024 would become moot and could not be implemented, as that pilot depends on the DAHRTP BEV rate."[10] Notably, Commission Rule 16.4(a) details that "A petition for modification asks the Commission to make changes to **an** issued decision" (emphasis added).[11] PG&E does not cite authorities or precedent to support its request to modify two Commission decisions, one explicitly and one implicitly, through a single PFM. The Commission should make an appropriate determination on the PFM considering this potential violation of Commission Rule 16.4(a).

Additionally, since D.22-10-024 was issued in October 2022 and the PFM was filed on February 5, 2025, PG&E has failed to comply with Rule 16.4(d). Although

---

[9] PFM at 3.
[10] PFM at 26.
[11] CPUC Rules of Practice and Procedure. Rule 16.4(a). https://www.cpuc.ca.gov/-/media/cpuc-website/divisions/administrative-law-judge-division/documents/rules-of-practice-and-procedure-may-2021.pdf

PG&E offers justification for filing the PFM more than one year after the effective date of D.21-11-017, it does not provide this justification in reference to D.22-10-024.

e. <u>The PFM misrepresents the state of PG&E's Vehicle-Grid Integration ("VGI") Pilots.</u>

PG&E argues that Phase 2 of the VGI Pilots offers an adequate near-term alternative to DAHRTP-BEV implementation for both unidirectional and bidirectional charging customers. However, as detailed above in section II.b., the PFM critically fails to note that the V2X Pilots are operating under budget limitations, including for program administration and equipment rebates, which will likely mean the V2X Pilot will, at some point soon, come to an end. Specifically, PG&E's website notes there are only incentives remaining for 126 chargers.[12] Given the mounting ratepayer affordability crisis (i.e., 47% residential electricity rate increase from 2019 through 2023[13]) and Executive Order N-5-24 on the rate crisis,[14] directing the Commission to do everything in its power to address it, VGIC acknowledges and accepts the high likelihood that PG&E's V2X Pilots will not receive additional funding beyond what has already been authorized. With this in mind, while the V2X Pilots provide an existing pathway for BEV customer participation in RTP, VGIC believes PG&E has misrepresented the reliability and sustainability of this interim solution and, as such, the argument that V2X Pilots provides an adequate alternative for DAHRTP-BEV should be disregarded by the Commission.

---

[12] PG&E. *Vehicle-to-Everything (V2X) pilot program.* https://www.pge.com/en/clean-energy/electric-vehicles/getting-started-with-electric-vehicles/vehicle-to-everything-v2x-pilot-programs.html. (Accessed March 7, 2025).

[13] Legislative Analyst's Office. *Assessing California's Climate Policies – Residential Electricity Rates in California.* January 7, 2025. https://lao.ca.gov/Publications/Report/4950

[14] Governor Gavin Newsom. *Executive Order N-5-24*. October 30, 2024. https://www.gov.ca.gov/wp-content/uploads/2024/10/energy-EO-10-30-24.pdf

f. <u>The PFM misrepresents BEV Parties' positions.</u>

VGIC commends PG&E and the other BEV parties for the consistently constructive and good faith collaboration that has underpinned D.21-11-017, D.22-10-024, and the proactive engagement since Q4 2023 as detailed in Declaration of Oriana Tell. All relevant parties have engaged in creative problem solving and sought customer-centric solutions in the face of ever-changing dynamics and challenges. That said, to ensure a factually correct record, VGIC notes that PG&E's claim that "The BEV parties, however, reserved their positions," in reference to the required relief discussed in 2024.[15] VGIC shared its position on the potential PFM relief request in a detailed response circulated to BEV parties on October 17, 2024. The position document is included as Attachment B at the bottom of this response.

III. THE COMMISSION SHOULD DENY THE PFM AS FILED BECAUSE THE REQUESTED RELIEF UNDERMINES THE COMMISSION'S OBJECTIVES AND PRINCIPLES AND WOULD MATERIALLY HARM PG&E RATEPAYERS.

a. <u>The PFM would materially harm customers and solution providers working to unlock load flexibility and exports to support California's grid.</u>

Approving the PFM as filed would limit opportunities for prospective customers to the VGI RTP rate implemented through PG&E's V2X Pilot. As detailed above in section II.b., this restricts the customer pool by imposing an additional requirement for customers that do not wish to participate through a PG&E-approved ASP. Additionally, for bidirectional charging customers, only customers with equipment listed on the "Eligible Product List for the V2X Commercial Pilot" may participate in the V2X Pilot, despite the availability of several additional vehicles and chargers that are currently capable of offering bidirectional charging services.

[15] PFM at 12. See also Declaration of Oriana Tiell, paragraph 13.

Additionally, as detailed immediately below, the PFM would effectively limit opportunities for dynamic load flexibility and V2G export to the CalFUSE framework, which faces critical headwinds including (1) the fundamental rate design and (2) the immense education gap for both solution providers and customers.

b. **The Commission has staked the future of load flexibility and distributed exports on a single strategy, CalFUSE, for which available data suggests there is little customer and vendor interest and limited load shift results.**

To be clear, **VGIC supports the CalFUSE framework**. It offers a hybrid dynamic pricing and demand response approach that has the potential to unlock load flexibility from certain customers and technologies. However, VGIC is concerned over the Commission's overreliance on the CalFUSE framework as the only strategy for unlocking load flexibility and V2G exports. VGIC notes two anecdotal data points that challenge CalFUSE's ability to reach all available customers interested in providing demand flexibility and/or V2G exports, and we cite two other data points that further highlights the design challenges inherent to the current CalFUSE framework.

First, the presence of the subscription component severely mutes the volatility of the real-time grid conditions, which fundamentally challenges the prospect of responding to real-time grid conditions. In practice, the subscription component, "frontloads" much of the value of participation into the first year of participation, especially for new V2G customers that are demonstrating combined load reduction and exports that far exceed the average EV customer baseline that would be applied. In the following years, the opportunity to save money is much smaller due to the presence of the baseline. In contrast, the DAHRTP BEV approach provides consistent opportunities to optimize charging and, for non-NEM export pilot participants, discharging around real-time grid conditions.

Second, the CalFUSE framework, in large part due to the subscription component and its scaling factors, presents an education gap for third-party providers and customers alike. VGIC makes no assertions that CalFUSE is too complex for all customers and vendors to understand, and we believe such an assertion would be paternalistic and would underestimate the intelligence of these individuals. However, we are concerned that the pace of the customer journey and the limited amount of time customers spend thinking about electricity is fundamentally at odds with the lengthy, complicated explanations required to explain the CalFUSE framework, including the exceptionally complex subscription component and its scaling factors.

VGIC notes that PG&E's VGI RTP has 65 customers (SAIDs) enrolled as of December 1, 2024.[16] PG&E does not disclose what share of these customers are residential or BEV, nor do they disclose what share are bidirectional (and receiving additional rebates to participate through the V2X Pilot upfront rebates) or unidirectional. If we were to assume all 65 VGI RTP customers are BEV customers, and consider that PG&E has 937 total BEV customers, the enrollment rate is just under 7%. Considering very few commercial EV charging stations are non-networked, and that customers "can't lose" due to the V2X Pilots offering only credits for performance and no penalties if customers perform worse than the OAT, VGIC believes this enrollment rate is very far below its potential.

Additionally, VGIC notes that initial data from Southern California Edison's ("SCE") CalFUSE Dynamic Rate Pilot, the only CalFUSE implementation open to EV customers that has completed robust evaluation, indicate that the structure has yet to

[16] Declaration of Oriana Tiell at paragraph 14.

create significant load shifting. SCE's recent final evaluation report notes that in the pilot, "the majority of the customers (14 out of 22) did not show evidence of a lower peak usage share," and any peak load reduction that was achieved was likely due to external factors.[17] This is very likely in part due to the subscription charges, which limits customers' exposure to the real-time prices. For customers that already have relatively low energy consumption baselines, there weren't many opportunities to capture meaningful bill savings. During high priced hours and days, customers did not face high real-time prices unless they consumed more than their subscription amount.

Taken together, this evidence suggests that the CalFUSE framework, though promising, is not well suited for all customers as it can fall apart due to a lack of education opportunities, the limited value due to the subscription component, and the apparent limits on resulting load shift, as suggested in the third-party analysis conducted on SCE's Dynamic Rate Pilot. DAHRTP BEV provides a meaningful alternative to CalFUSE and allows customers, PG&E, and all stakeholders to learn about how customers respond to more exposure to RTP signals. By being exposed to more meaningful price signals, DAHRTP customers will be further incentivized to shift their load and export to lower priced hours or days, and VGIC anticipates that more significant load shifting and power export is likely under this structure. This is especially true for commercial BEV customers that face very high electricity costs for their EV charging, as these customers are more likely to be working with third party solution providers, which may include PG&E-approved ASPs but may include other sophisticated algorithm-based technologies, that are working to optimize BEV

[17] SCE. *Final Evaluation Report for Southern California Edison's Dynamic Rate Pilot.* Circulated February 28, 2025 to service lists of R.20-11-003 and R.22-07-005.

charging using software tools. If these customers are able to better capture bill savings and avoid high prices, they will be more inclined to take that option.

It is also important that the Commission pursue a wide variety of load shifting strategies for EVs at this time because charging needs are so diverse, especially for non-residential customers. These customers will have very different power needs for their EVSE, from less than 50 kW to over 1,000 kW, and different charging schedules, with some vehicle remaining parked for significant amounts of time while others need fast charging to be able to continue driving along a route. Given the considerable diversity of charging needs across consumers, it will be most beneficial for customers to be given a wide range of charging schedules, rates, and tools to manage charging to find what best fits their specific needs and internal cost-benefit calculations. Offering a variety of options can enhance public acceptability and feasibility by tailoring approaches to the diverse needs of customers and stakeholders. DAHRTP is an important option to pursue as part of that flexibility.

c. Over-relying on a single approach is at odds with established energy planning principles of resource diversity and supplier diversity, strategies intended to mitigate risk.

Offering both CalFUSE and DAHRTP BEV options to customers is consistent with California's well-established approach to resource diversity and supplier diversity. Just as the Commission orders and authorizes the utilities to construct or contract with a range of centralized electric generation resources, including onshore and offshore wind, utility-scale solar, and utility-scale energy storage, the Commission must promote resource diversity in its distributed energy resources. Similarly, just as utilities rely on a range of suppliers for its service vehicles, equipment, and technology providers, utilities must promote supplier diversity as it sources demand side flexibility and distributed energy resource exports. These diversification strategies mitigate risk and, especially in a challenging business environment for startups and emerging technologies, provides protections against any individual actors exercising market power and resilience against solution providers going out of business, leaving behind stranded assets.

Additionally, the over-reliance on CalFUSE carries a risk of "path dependence" – the influence of past decisions on future outcomes, hindering the implementation of alternative solutions. In the case of VGI customers, VGIC is concerned that relying too heavily on a single dynamic rate approach for all customers, as PG&E implies in its PFM to be the CalFUSE framework, would lock in a specific future that may not, in fact, be in the best interest of ratepayers. Notably, Cal Advocates has published findings suggesting that optimized VGI implementation can reduce electrification-driven distribution system costs from the Commission's estimated $50 billion through 2035 down to $22.7 billion.[18] Given the magnitude of these numbers, the stakes are incredibly high, especially for low-income and other vulnerable California ratepayers already feeling an outsized energy cost burden. However, to achieve the reductions modeled by Cal Advocates, IOUs will need to engage EV customers at a significant enough scale for these optimized load shapes to have a beneficial impact on the total required distribution system upgrades. VGIC is concerned that pursuing only the CalFUSE framework, risks a gamble with path dependence, wherein PG&E would be

[18] *See* CPUC Electrification Impacts Study Part 1: https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M508/K423/508423247.PDF and Cal Advocates DGEM 2.0 Preliminary Results, October 21, 2024. https://www.publicadvocates.cpuc.ca.gov/-/media/cal-advocates-website/files/press-room/reports-and-analyses/241024-public-advocates-office-dgem-20-preliminary-results.pdf

"locked in" to offering only CalFUSE RTP rates, with no opportunity to understand EV customer interest in the DAHRTP BEV rate design until it is too late to avoid the upgrades forecasted by 2035 and the resulting ratepayer costs of up to $50 billion.

d. Approving the PFM would send a strong signal that regulated utilities can successfully delay and then terminate implementation efforts for Commission orders they do not wish to comply with.

As detailed in section I above, PG&E had planned to launch DAHRTP in October 2023 and later received approval for several extension request. The PFM details PG&E's assessment that they would be unable to implement DAHRTP until the completion of billing system upgrades in 2029. VGIC is concerned over the signal that this level of delay and resulting PFM sends to regulated entities and all stakeholders regarding other existing and future Commission orders.

If the Commission grants the PFM to cancel DAHRTP implementation, it will send a clear signal that Commission-regulated entities can elect to effectively nullify regulatory decisions through continued delays. Such a standard will erode confidence in the Commission's ability to regulate utilities effectively. Stakeholders and customers depend on the Commission to uphold its decisions and ensure timely execution of critical programs. This is particularly true for programs aiming to tap into emerging technologies and unlock a broader commercial ecosystem that will provide services to California for years and decades to come, such as the VGI market. If utilities observe that Commission orders can be sidestepped by citing system limitations or technical constraints, it will undermine the Commission's credibility and deter future policy progress. Moreover, this delay undermines the integrity of the public engagement process. The optional DAHRTP BEV rate and associated non-NEM export pilot were the results of extensive productive and good-faith stakeholder discussions, and, by allowing PG&E to avoid implementation, stakeholders who invested time and resources in shaping this policy will find it difficult to justify future investments in Commission initiatives.

To maintain trust in the Commission's decision-making process, the Commission should deny PG&E's PFM as filed and instead adopt the Alternative Proposed Modifications detailed below.

IV. VGIC PROPOSES ALTERNATIVE PROPOSED MODIFICATIONS FOR THE COMMISSION'S CONSIDERATION.

VGIC offers below three straightforward Alternative Proposed Modifications to the DAHRTP BEV approach that addresses PG&E's three primary reasons for submitting the PFM. We respectfully request the Commission consider these Alternative Proposed Modifications and offer our team as a resource to discuss each with the Commission, Energy Division staff, PG&E, and other parties. VGIC offers specific redlines to D.21-11-017 Ordering Paragraphs in Attachment A below its response.

a. **The Commission should direct PG&E to leverage its already completed work on dynamic distribution components to promote compliance with the California Energy Commission's Load Management Standard.**

PG&E notes throughout the PFM that the DAHRTP-BEV rate and associated export compensation pilot would not comply with the CEC's LMS due to the absence of a dynamic distribution component. VGIC strongly recommends PG&E include a dynamic distribution component in the DAHRTP-BEV rate and export compensation pilot and, fortunately, PG&E has **already completed the work to develop both a dynamic distribution rate methodology and**

communication platform. Through its Hourly Flex Pricing pilot and VGI RTP rate pilot, PG&E currently published hourly distribution price signals based on marginal distribution system costs for each customer's representative circuit. With this in mind, VGIC views a strong opportunity to simply adopt this dynamic distribution component into DAHRTP-BEV, thereby putting PG&E in compliance with the CEC LMS and enabling DAHRTP-BEV to represent a more complete value stack for customers and third-party solution providers.

b. The Commission should direct PG&E to enable shadow billing for initial DAHRTP-BEV implementation.

The PFM and associated Declarations note that PG&E explored implementing shadow billing for DAHRTP-BEV and determined that it was not feasible. While VGIC appreciates that implementing DAHRTP-BEV through a shadow bill presents a challenge, we find PG&E's argument as to why this is infeasible to be incomplete. The Declaration of Andrew Au states the following:

> "The DAHRTP BEV opt-in rate approved in D.21-11-017 cannot be shadow billed because it can produce RTP revenues that exceed the non-RTP BEV tariff OAT revenues. Shadow billing for the CalFUSE expanded pilots and VGI pilot was approved as a credit only program, *i.e.,* no additional customer payments if their shadow RTP bill is higher than their otherwise applicable tariff (OAT) bill."[19]

VGIC does not dispute that the shadow billing approach for the CalFUSE expanded pilots and the associated VGI pilot was approved as a credit-only program. However, the Commission has not adopted a definition of "shadow billing" that precludes its use outside of CalFUSE in a manner other than "credit-only," nor has PG&E detailed why, on a practical level, only credits can be issued through shadow

[19] Declaration of Andrew Au at paragraph 6.

billing. With this in mind, VGIC acknowledges that shadow billing for DAHRTP-BEV would not be identical to the shadow billing approach for CalFUSE. However, VGIC believes PG&E holds the expertise and capability to adapt shadow billing to be useful beyond the credit-only approach and, in turn, support DAHRTP-BEV. As such, VGIC urges the Commission to consider directing shadow billing be used to implement DAHRTP-BEV, albeit through a modified shadow billing methodology from what is in place for CalFUSE pilots limited to a credit-only approach.

c. **The Commission should authorize budget transfer within the already approved DAHRTP-BEV funds to support the Alternative Proposed Modifications.**

To support the modification recommended in the above, VGIC recommends the Commission authorize additional budget transfers within DAHRTP-BEV to support the establishment of a modified shadow billing approach. As PG&E has not detailed the budget amount it requires to implement shadow billing for DAHRTP-BEV customers, VGIC recommends the Commission direct PG&E to submit an Advice Letter to request approval for additional budget, as needed, to support modified shadow billing for implementation or DAHRTP-BEV and the associated export pilot.

V. CONCLUSION.

VGIC appreciates the opportunity to submit this response to the PFM. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen
Zach Woogen
Executive Director
VEHICLE-GRID INTEGRATION COUNCIL

March 7, 2025

Attachment A:

- OP 2:
    - Pacific Gas and Electric Company shall develop a marginal energy cost based on California Independent System Operator's day-ahead pricing and average default load aggregation point loss factor and a marginal distribution cost based on the marginal distribution cost calculated under Pacific Gas and Electric's Expanded Dynamic Rate Pilot and Vehicle Grid Integration Pilots authorized in D.24-01-032 and Resolution E-5192 for the optional day-ahead real time rate adopted in this proceeding and the export compensation pilot authorized in D.22-10-024.
- New OP:
    - Pacific Gas and Electric Company shall implement the DAHRTP rate authorized in this Decision for commercial electric vehicle customers and the export compensation pilot authorized in D.22-10-024 using a shadow billing approach until PG&E's billing system can accommodate the necessary updates to no longer require a shadow billing approach.
- OP 12:
    - Within 30 days of this issuance of this Decision, Pacific Gas and Electric Company shall file a Tier 1 advice letter creating a Dynamic and Real-Time Pricing Memorandum Account that includes a subaccount to specifically track costs associated with implementing the day-ahead, real-time hourly pricing (DAHRTP) rate authorized in this Decision for commercial electric vehicle customers. PG&E is authorized to recover up to $6 million to implement its optional DAHRTP rate including any costs incurred to develop and implement a shadow billing approach to enable customer participation. If, 24 months after the issuance of this Decision, the $6 million budget cannot support additional customer enrollment in the DAHRTP rate, PG&E may file a Tier 2 advice letter seeking authorization to recover up to an additional $3.6 million to continue enrolling and evaluating customers' responses to the DAHRTP rate.

Attachment B:

<u>VGIC Feedback on October DAHRTP Drafts</u>

October 31, 2024

VGIC opposes the redlined settlement agreement and would not support the draft PFM unless modified as recommended below.

1. PG&E's Primary Reason for Filing the PFM, Amended Settlement, and Redlined Settlement Hinges on DAHRTP BEV Lacking a Dynamic Distribution Price Signal, Thereby Making it "Obsolete" and Non-Compliant with CEC LMS.
   A. **VGIC Proposes the PFM and Settlement be Modified to Adopt the Distribution Price Signal Currently Implemented in VGI Pilot and Expanded Dynamic Rate Pilots.**

2. PG&E's Secondary Reason for Filing the PFM, Amended Settlement, and Redlined Settlement Hinges on BCS Deployment Preventing DAHRTP BEV From Launching.
   A. **VGIC Proposes the Settlement be Redlined to Allow PG&E to "Shadow Bill" the DAHRTP BEV Rate Until BCS Deployment is Complete.**

3. PG&E's approach critically lacks an uninterrupted solution for both BEV V1G and BEV V2G customers. The plan to offer BEV customers (V1G and V2G) participation in HFP via the VGI Pilots until the pilot funding expires and then offer these customers 2023 GRC II HFP rates does not square with the timelines of these respective developments. The Commercial VGI Pilot is 74% subscribed, which suggests that the funding may not last much longer. Given the mounting ratepayer affordability crisis, the outcome in the most recent CPUC Resolution on the VGI Pilots (which did not fulfill PG&E's entire funding request), and yesterday's Executive Order on the rate crisis, directing the CPUC to do everything in its power to address it, settlement parties must collectively acknowledge and accept the high likelihood that PG&E's VGI Pilots will not receive additional funding beyond what has already been authorized. Meanwhile, the 2023 GRC II HFP rates will not be implemented until at least mid-2026, which PG&E acknowledges in its 2023 GRC II testimony.
   A. VGIC's singular condition since the second delay came to light in December 2023, shared **consistently and clearly** during calls with settling parties throughout 2024, has been to ensure that there is continuity and a clear path forward for participating customers despite the BCS update delays. We have also clearly communicated that this condition can and should be addressed via written commitment in PG&E's Commission filings detailing this path forward for these customers. However, the approach shared in these documents **makes no effort to address this singular condition** or even reference the challenge this poses to customers. It is, therefore, more important than ever that PG&E follow through on its settlement agreements to implement a full-scale, optional dynamic rate and companion export compensation. VGIC has been inspired by PG&E's commitment to large-scale VGI ranging from Patti Pope's public remarks, to setting ambitious internal VGI goals, to investing in a team of highly-

skilled, passionate, and motivated VGI program designers and implementers that VGIC staff and members have had the pleasure of collaborating with and learning from. VGIC has held the honor of appearing alongside PG&E representatives in several virtual and in-person public forums where both parties promoted PG&E's leadership and commitment in the space. However, in agreeing to the redlined settlement and supporting the PFM, VGIC would directly betray its mission and we believe that PG&E, too, in taking these steps, will inexplicably and inexcusably obstruct its own clearly-stated VGI goals.

4. **Initial VGI Pilot Phase 2 Insights:** Recent insights from VGI industry participants suggest that PG&E's VGI Pilots place a great deal of limitation on customers and, in turn, successful implementation for BEV customers (both V1G and V2G) within the VGI Pilots structure is increasingly unlikely:
   A. **HFP Rate Design**: Until recently, companies were unable to properly model the HFP approach due to the lack of a clear, published methodology or tools for customers/vendors to calculate the distribution price signal. While PG&E provided VGIC with its estimates for an average BEV customer on DAHRTP (generation, no distribution, no baseline) vs HFP (generation, distribution, baseline), the recent publication of the HFP signals on the HFP website has made clear that the rate will not financially support projects in the long-term.
      - Specifically, the implementation of the subscription element undercuts the financial viability of bidirectional charger and charger-paired BESS deployment. This is because customers would effectively be "competing" against their prior year's load shape. In effect, the HFP subscription element has recreated the perennial baselining issue that has limited EV participation in demand response programs for years, but simply placed it on a different time scale (year-prior, as opposed to 10-in-10). The subscription element severely mutes the HFP signal and, as a result, will not meaningfully spur sustained participation and investment from ASPs. This fundamentally puts BEV participation at risk because, having administered the VGI Pilots, PG&E knows better than most that customer participation, especially from bidirectional charging customers, cannot be unlocked without ASPs. **Simply put, bidirectional charging customers have no ability to discharge their vehicle batteries without these technology partners.** There is no "export now" button on a vehicle, charger, or smartphone app, as that is fundamentally not how these products are being designed for BEV customers or, for that matter, residential customers driving light-duty vehicles.
      - In contrast, the DAHRTP approach detailed in the settlement agreement and adopted across three separate Commission orders spanning two years of record does not incorporate this limiting subscription component and, instead, exposes participating customers to real-time-equivalent marginal generation costs (i.e., for energy and generation capacity). This means the volatility is not muted and, in turn, supports more meaningful customer revenues and a viable, sustainable shared value proposition between participants, ASPs, PG&E, and non-participating ratepayers.

   B. **Approved equipment and ASP/aggregator requirements:** Customers participating in the VGI Pilots must do so with PG&E-approved equipment and through an aggregator that has an agreement with an automotive OEM. In contrast, the DAHRTP export settlement agreement does not impose any such requirements. Of course, customer-sited generation and energy storage equipment must comply with PG&E's Rule 21 interconnection requirements.

However, the DAHRTP export settlement agreement supports an open market where technology providers can compete based on customer service and equipment price, rather than artificially limiting participation to owners with certain equipment participating through certain types of ASPs/aggregators. This is evident today, where PG&E's V2X Pilots webpage, lists only one eligible charger and vehicle for the residential pilot, and only one EV manufacturer and one charger manufacturer for the commercial pilot. However, VGIC is aware of the following UL-certified bidirectional chargers that are not deemed eligible within PG&E's VGI Pilots:

- Commercial
  - BorgWarner 60 kW
  - BorgWarner 125 kW
  - Fermata FE-15 15 kW
  - Fermata FE-20 20 kW
  - InCharge US 22 kW
- Residential
  - GM Energy V2X Charger
  - Tesla Wall Connector + Tesla Gateway

5. **Implications of Billing System Upgrade Delays:** During the settlement discussions, the billing system updates were not refuted by any party. VGIC felt settling parties were operating under an implicit, collective understanding that PG&E was making an informed accounting of the timeline required to update its own billing system. **PG&E, unchallenged by settling parties, significantly underestimated its BCS update needs.** VGIC sympathizes with the challenges of estimating billing system upgrade costs and understands that the reality of billing system implementation relies on a patchwork of suboptimal temporary fixes that, through dedicated technical expertise and innovative problem solving, manage to meet the challenging, high-stakes task of customer billing. VGIC also understands the inherent risks posed to non-participating customers when billing systems are being modernized. *However,* PG&E must not shift the risk it accepted by estimating these billing system updates onto customers in the form of rescinding this important VGI program opportunity. In other words, PG&E's unforced error must not become BEV customers' barrier to achieving lower costs of ownership or ratepayers' burden of forgone avoided system costs. Fortunately, PG&E has implemented a viable alternative for other RTP approaches, shadow billing, which should be leveraged for BEV DARHTP customers, as well.
6. **Marketing:** Under the interim approach of implementing BEV RTP-equivalent pricing via the V2X Pilots, customers are not receiving adequate education and marketing to participate in the rate. PG&E's "EV rate plans," "Electric Vehicles," and "Rate plans" webpages do not provide any information or links to this option. PG&E's "Hourly Flex Pricing" webpage does contain the relevant information, however, based on VGIC's navigation (conducted October 29, 2024), there is no pathway for customers to reach this webpage through the three relevant webpages noted above, or through any other means other than Google. For those seeking participation in HFP based on word of mouth, Google may be sufficient. For those seeking participation in HFP through an Automated Service Provider (ASP) or through PG&E's Fleet Advisory Services Program (although PG&E has not detailed that it is sharing information about HFP within its program), an easily accessible webpage is not necessarily critical. However, for customers outside of ASP, word-of-mouth, and Fleet Advisory Services, PG&E does not link to its HFP webpage from all relevant webpages (i.e., "EV rate plans," "Electric Vehicles," and "Rate plans"). This further demonstrates that practical differences between a time- and funds-limited PG&E-administered pilot and a full-scale, optional

rate, which would necessarily appear in its rate webpage and tariffs book and on its "Rate plans" and "EV rate plans" web pages alongside all other relevant EV rate plans.

7. **VGIC Recommended Path Forward:** VGIC recommends settling parties embrace the following two changes to address the concerns raised in PG&E's PFM:
   A. The settlement agreement should be redlined and re-signed by settling parties to incorporate a dynamic distribution rate component based on the same marginal cost-based methodology implemented in PG&E's other RTP rates (HFP and VGI Pilot HFP).
   B. The settlement agreement should be redlined and re-signed by settling parties to adopt shadow billing for BEV DAHRTP and BEV DAHRTP Non-NEM Export customers until the billing modernization updates are complete.